TOGAF Content Metamodel

Module 7

Module Objectives

- What a metamodel is and why it is needed
- Key concepts of the Core Metamodel
- The division of the metamodel into Core and Extensions
- Key concepts of the Core Metamodel Entitites
- The components of the TOGAF Content Metamodel

What is a metamodel?

- A metamodel is a **precise definition** of the constructs and rules needed for creating models
- A model that **describes how** and **with what** the architecture will be described in a structured way

Benefits of the Metamodel

- It formalizes **the definition** of an Enterprise Architecture
- It formalizes the relationships between objects
- It enables an **EA tool mapping**

Formal and Informal Modeling

- When defining architecture content there are choices to be made on the level of **structure** and **formality**
- In some cases very formal specific language is needed in order to articulate and govern in a precise or detailed way
- In other cases the use of formal engineering discipline will result in architecture content that is
  - inappropriate for the audience
  - difficult to communicate

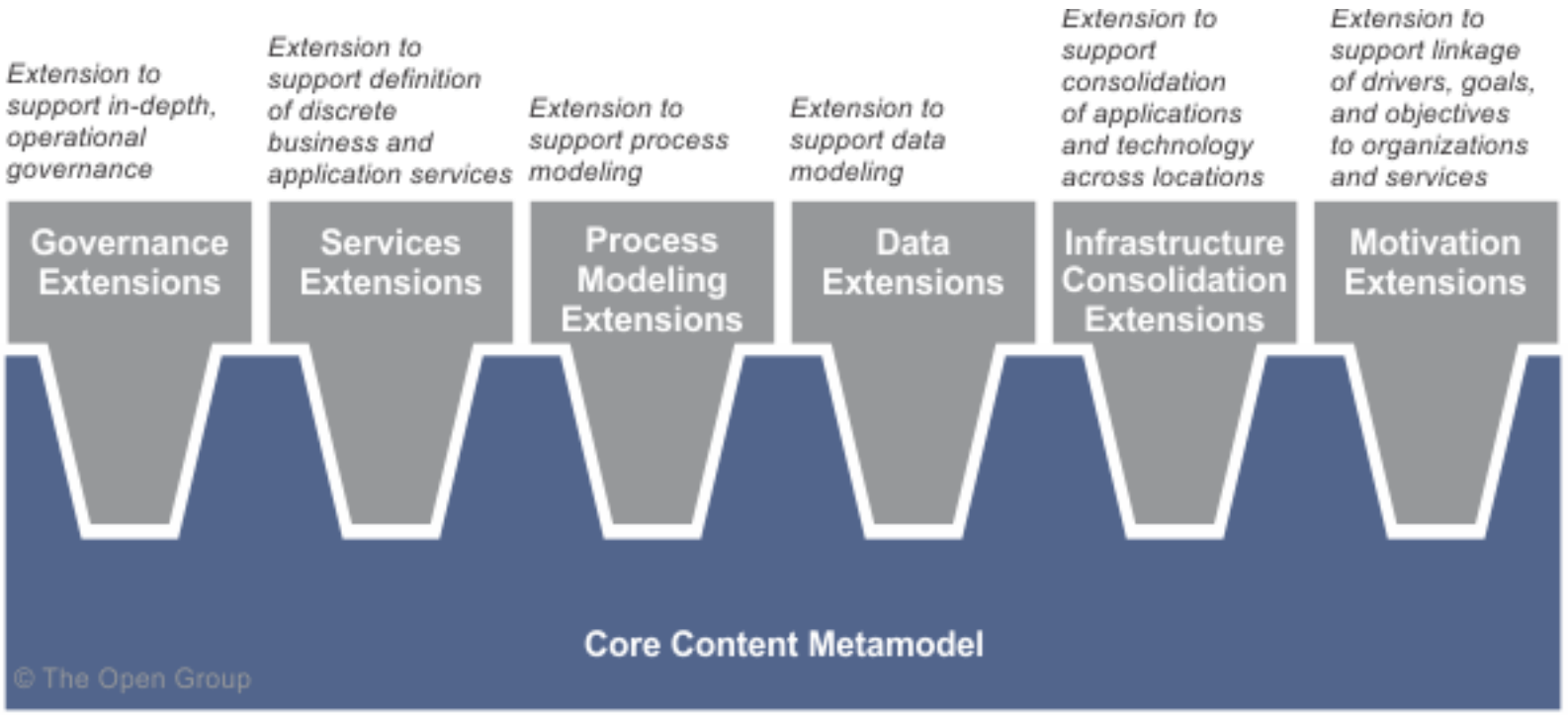
Core Content Metamodel COncepts

- A TOGAF architecture is based on
  - Defining **architectural building blocks** within architecture catalogs
  - Specifying the **relationships** between those building blocks in architecture matrices
  - And presenting **communication diagrams** that show in a precise way what the architecture is
- The metamodel is structured into **Core** and **Extension** content
  - Core contents is designed not to be altered

Core and Extension Content

- In order to support many scenarios the metamodel has been partitioned into **core** and **extension** content
- The **core** provides a minimum set of architectural content to support traceability across artifacts
- The **extension** content allows for more specific or more in depth modeling

TOGAF Content Metamodel and its Extensions



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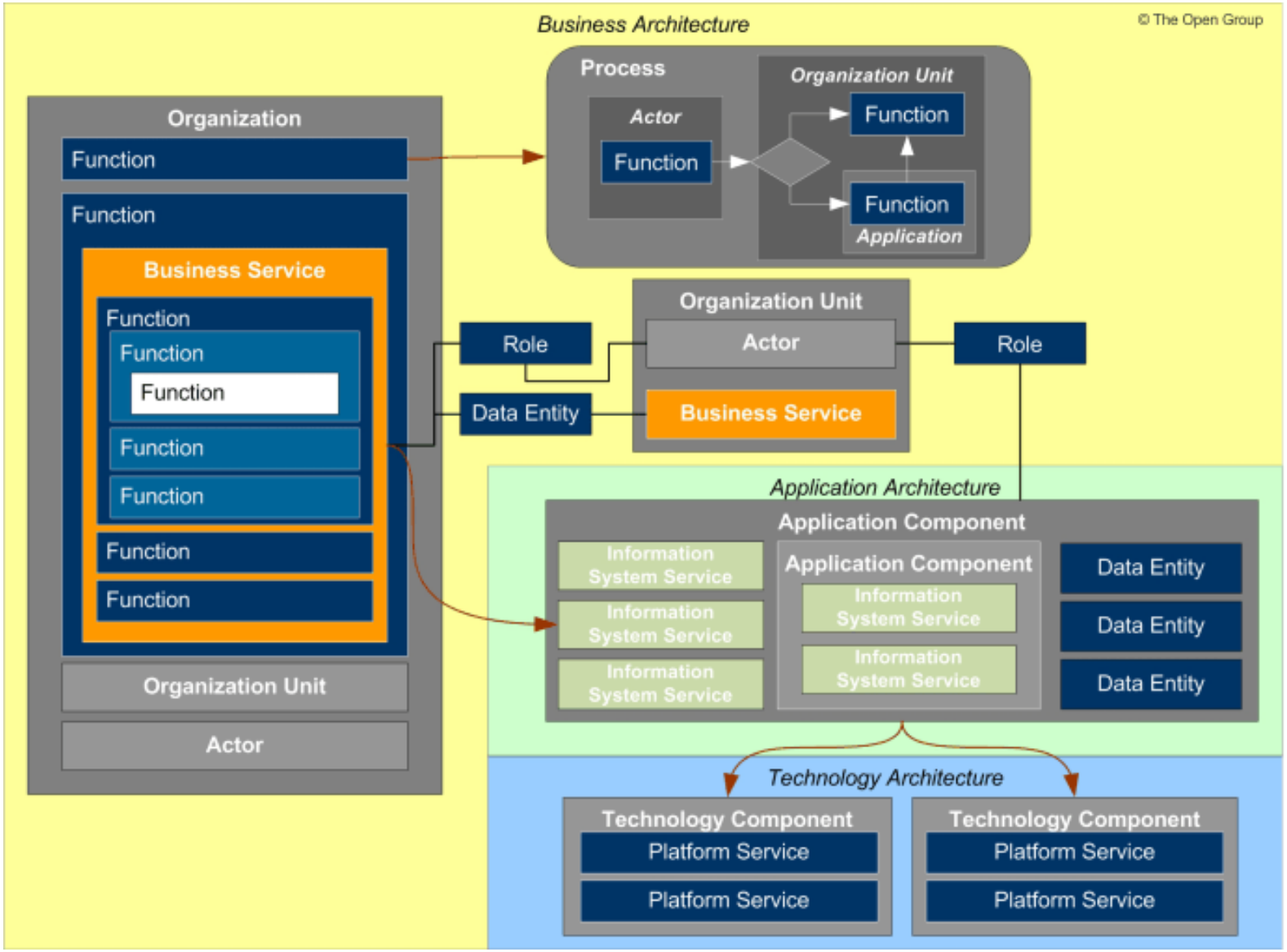
Core Metamodel Entities

- **Actor:** A person, organization or system that is outside the consideration of the architecture model, but interacts with it
- **Application Component:** An encapsulation of application functionality that is aligned to implementation structuring
- **Business Service:** Supports business capabilities through an explicitly defined interface and is explicitly governed by an organization
- **Data Entity:** An encapsulation of data that is recognized by a business domain expert as a discrete concept. Data entities can be tied to applications, repositories, and services and may be structured according to implementation considerations
- **Function:** Delivers business capabilities closely aligned to an organization, but not explicitly governed by the organization

Core Metamodel Entities (Cont'd)

- **Information System Service:** The automated elements of a business service. It may deliver or support all of one or more business services
- **Organization Unit:** A self-contained unit of resources with line management responsibility, goals, objectives and measures. Organization units may include external parties and business partner organizations
- **Platform Service:** technical capability required to provide enabling infrastructure that supports the delivery of applications
- **Role:** An actor assumes a role to perform a task
- **Technology Component:** An encapsulation of technology infrastructure that represents a class of technology product or specific technology product

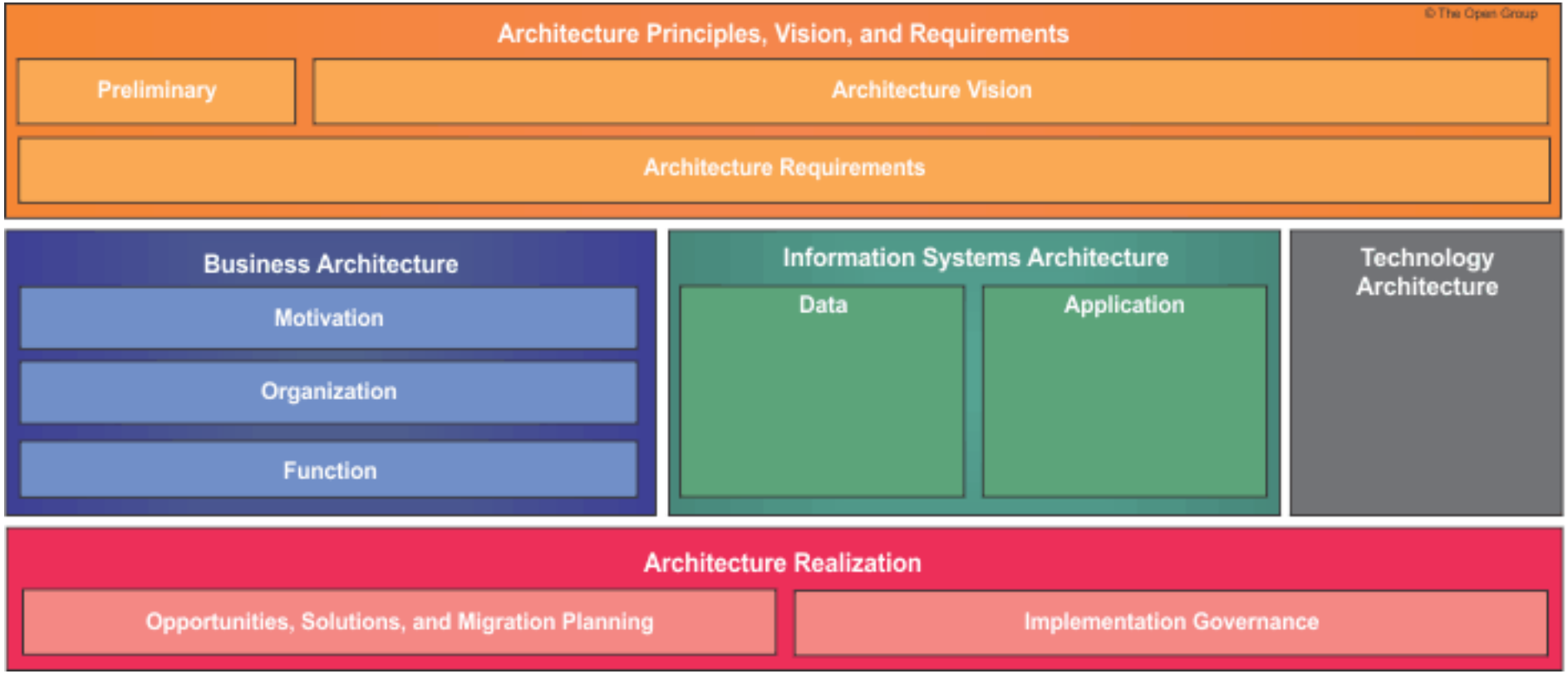
Core Entities and their Relationships



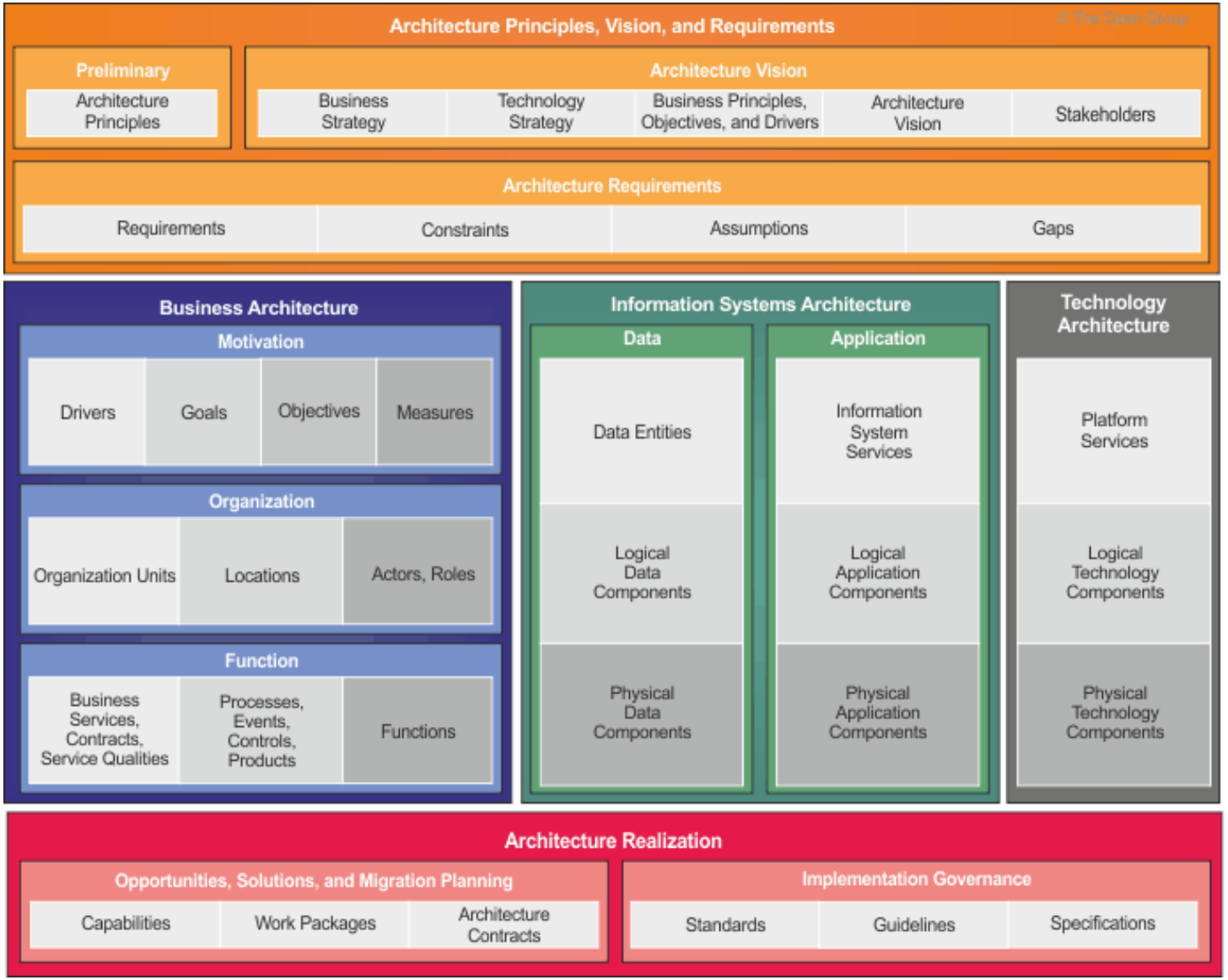
The Content Metamodel

- The content metamodel **provides definitions of all the types** of building blocks that may exist, showing how they can be described and related to one another
- When creating and managing architectures, it is necessary to consider **concerns** such as business services, actors, applications, data entities, and technology
- The metamodel highlights these concerns, shows their relationships and **identifies artifacts** that can be used to represent them in a consistent way
- The metamodel can also be used to **provide guidance** to organizations that wish to implement their architecture using an architecture tool

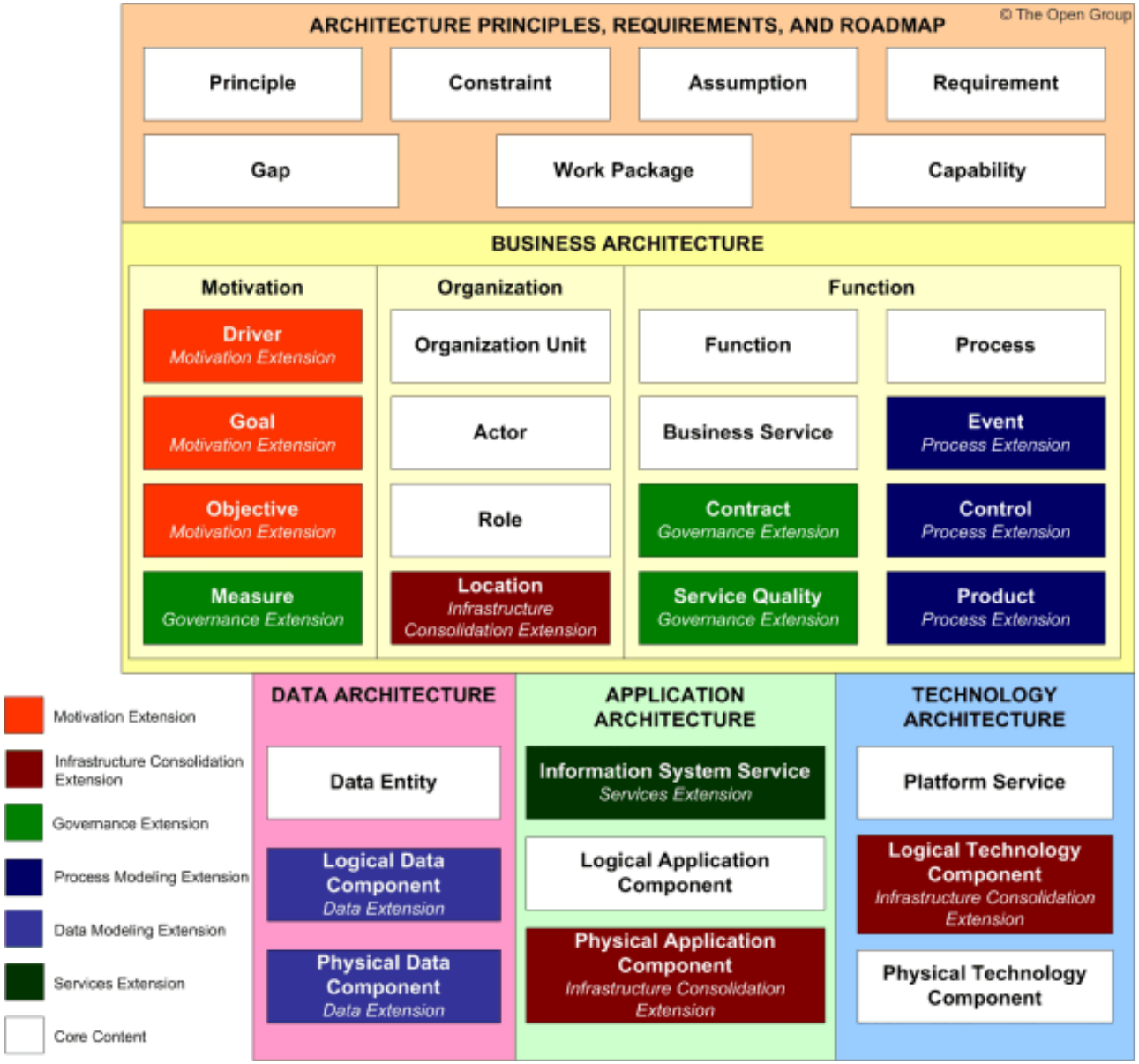
Content Metamodel (Simplified)



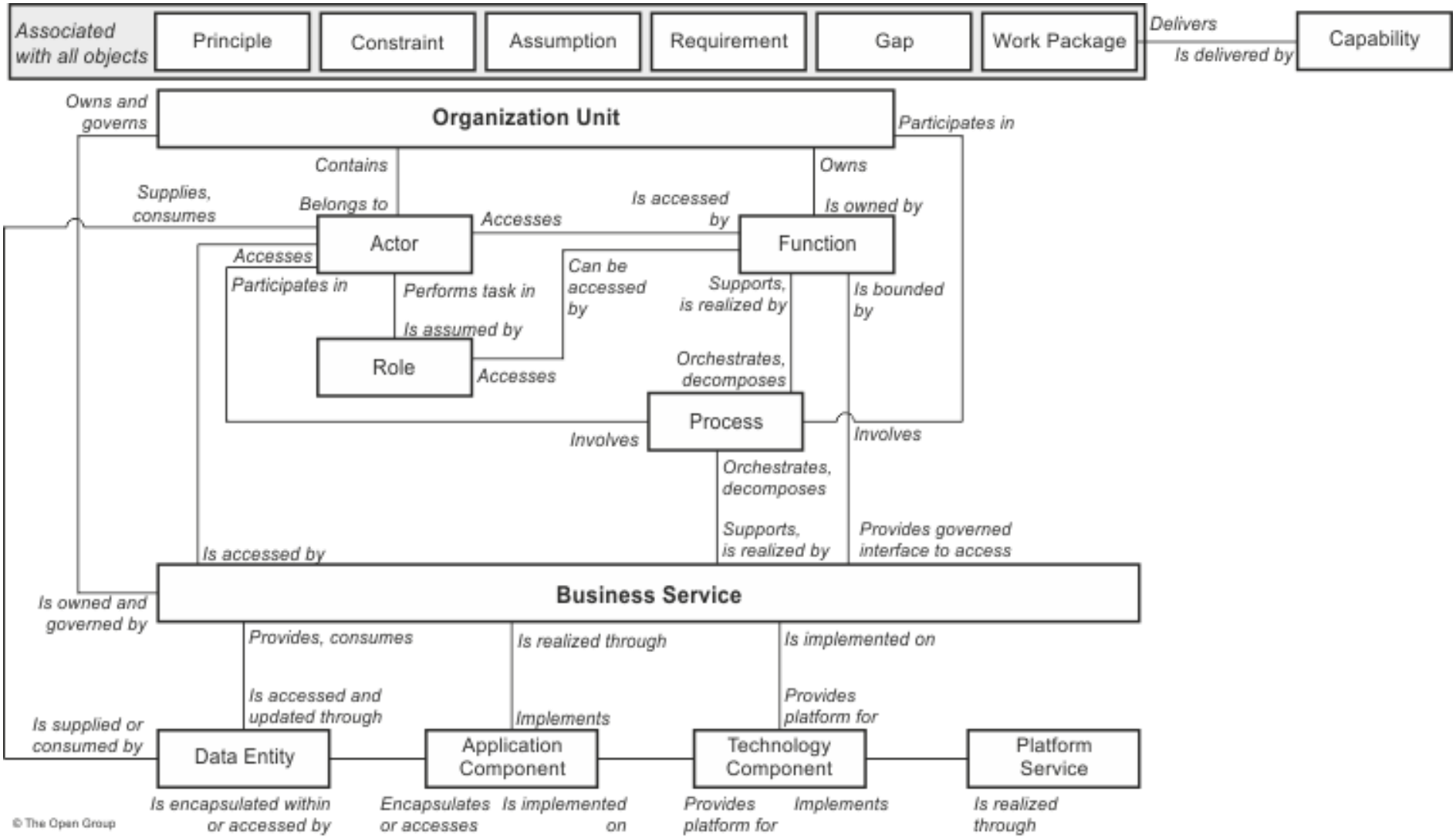
Content Metamodel (Detailed)



Full Content Metamodel



Core Content Metamodel



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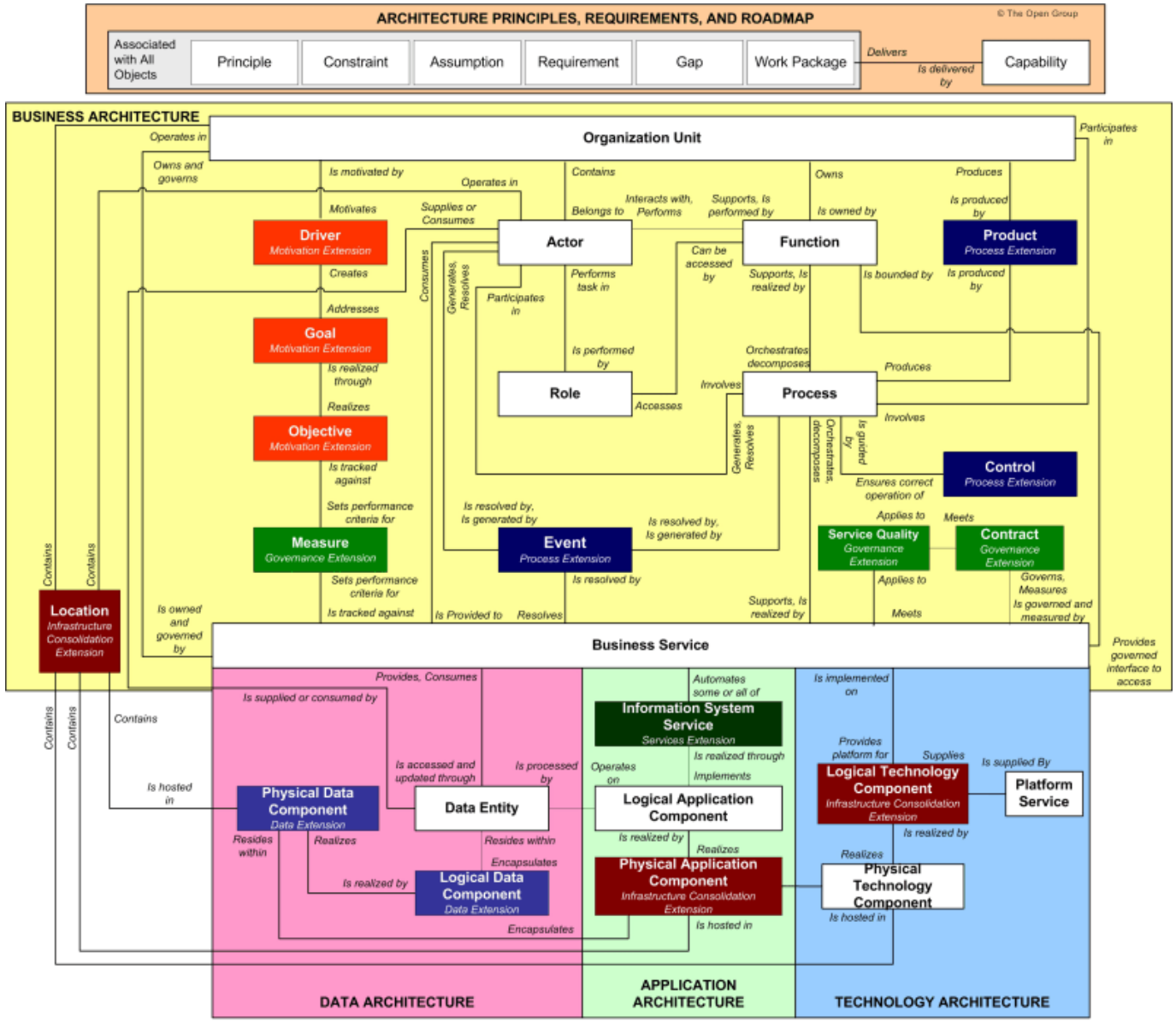
Core TOGAF 9 Artifacts

<table>
<tr><td>Preliminary Phase
• Principles catalog</td><td colspan="3">Phase A, Architecture Vision
• Stakeholder Map Matrix • Solution Concept diagram • Value Chain diagram</td></tr>
<tr><td colspan="4">Requirements Management • Requirements catalog</td></tr>
<tr><td>Phase B, Business Architecture
• Organization/Actor catalog
• Driver/Goal/Objective catalog
• Role catalog
• Business Service/Function catalog
• Location catalog
• Process/Event/Control/Product catalog
• Contract/Measure catalog
• Business Interaction matrix
• Actor/Role matrix
• Business Footprint diagram
• Business Service/Information diagram
• Functional Decomposition diagram
• Product Lifecycle diagram</td><td>Phase C, Data Architecture
• Data Entity/Data Component catalog
• Data Entity/Business Function matrix
• Application/Data matrix
• Logical Data diagram
• Data Dissemination diagram</td><td>Phase C, Application Architecture
• Application Portfolio catalog
• Interface catalog
• Application/Organization matrix
• Role/Application matrix
• Application/Function matrix
• Application Interaction matrix
• Application Communication diagram
• Application and User Location diagram
• Application Use-Case diagram</td><td>Phase D, Technology Architecture
• Technology Standards catalog
• Technology Portfolio catalog
• System/Technology matrix
• Environments and Locations diagram
• Platform Decomposition diagram</td></tr>
<tr><td colspan="4">Phase E. Opportunities & Solutions
• Project Context diagram
• Benefits diagram

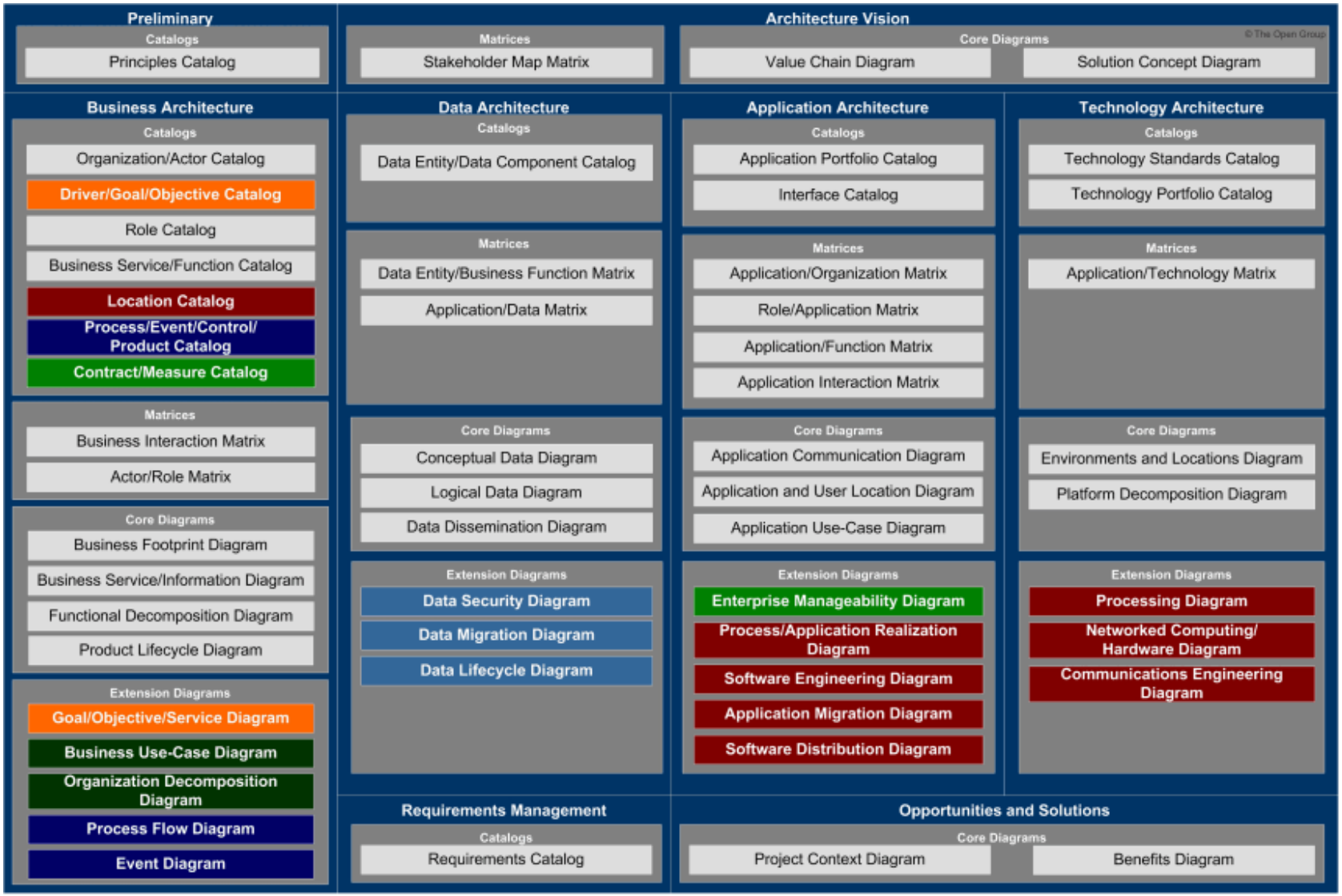
Core TOGAF 9 Artifacts</td></tr>
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Full Content Metamodel with Relationships

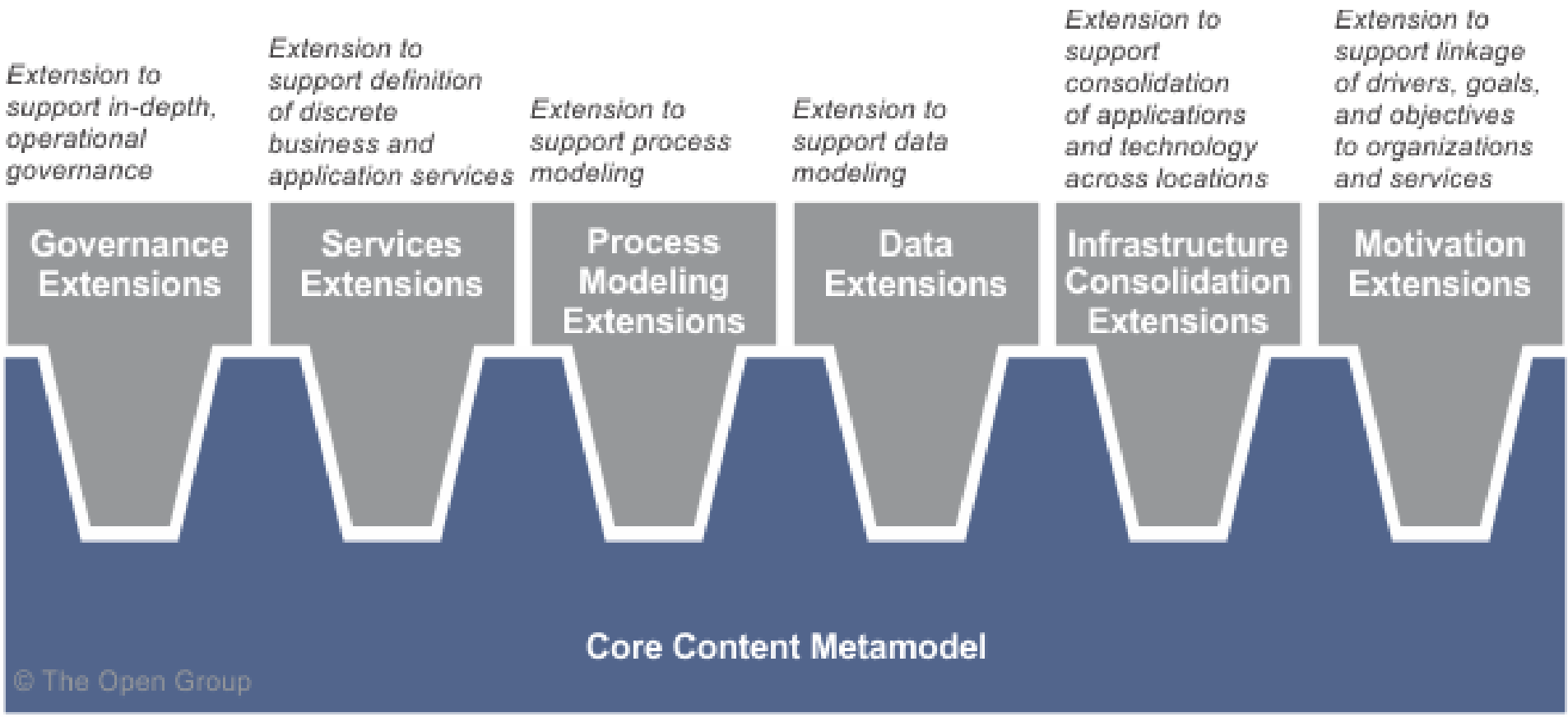


Full Content Metamodel Artifacts

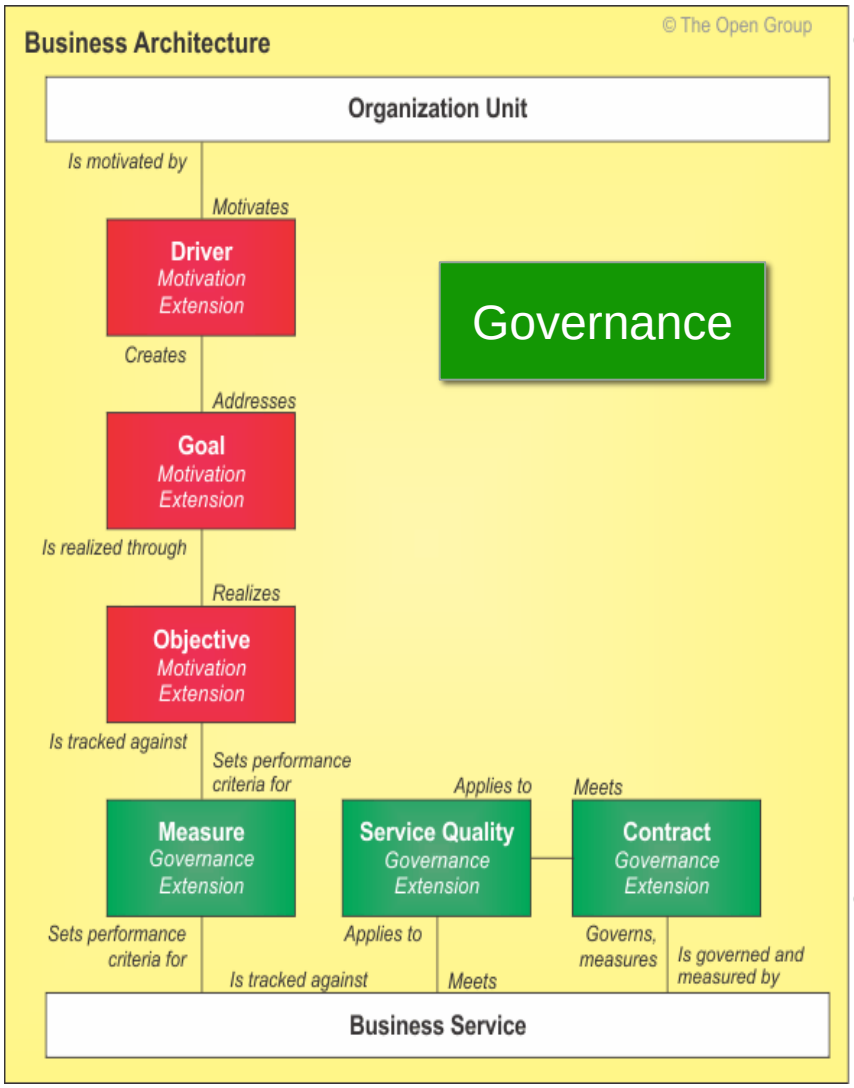


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Metamodel Extensions Overview

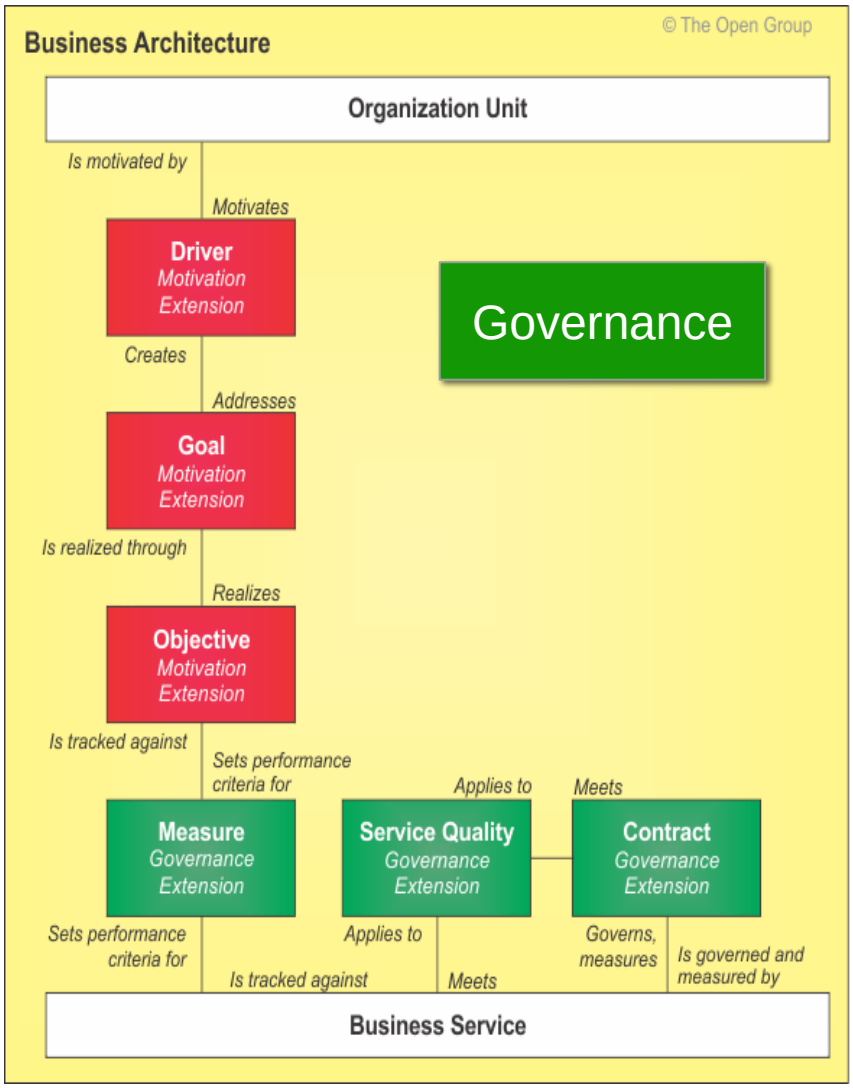


Governance Extension



- Scope
  - The ability to apply **measures to objectives** and then link those **measures to services**
  - The ability to apply **contracts to service communication** or service interactions with external users and systems
  - The ability to define **re-usable service qualities** defining a service-level profile that can be used in contracts
  - Creating of additional diagrams to show ownership and management of systems
- Additional diagrams to be created
  - **Enterprise Manageability Diagram**

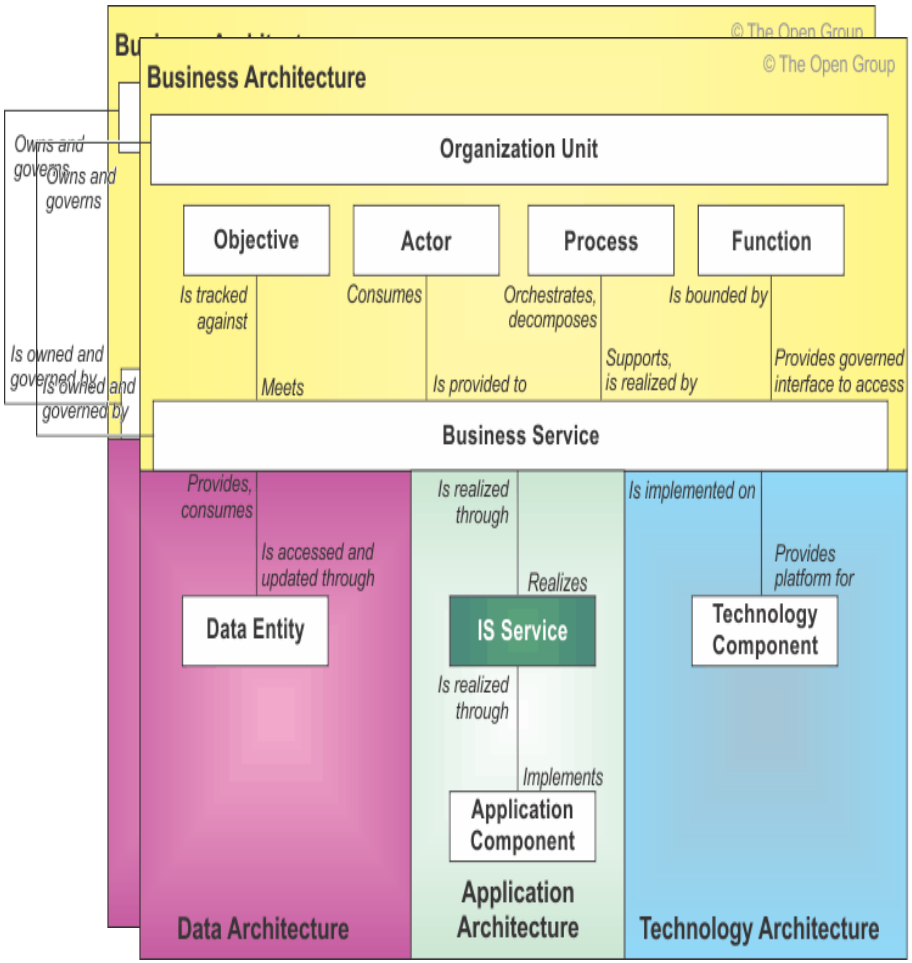
Governance Extension



- Should be used
  - When an organization is considering **IT change** that will result in a significant impact to existing operational governance models
  - When an organization has **granular requirements** for service levels that differ from service to service
  - When an organization is looking to **transform its operational governance** practice

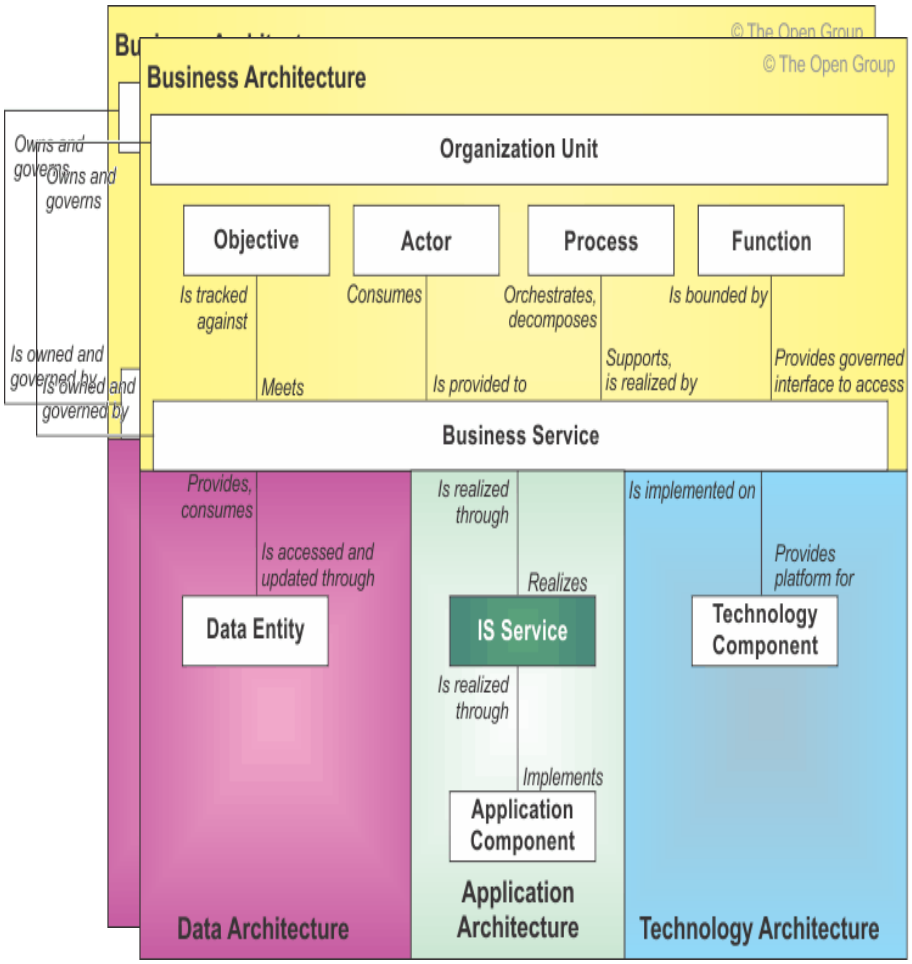
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Services Extension



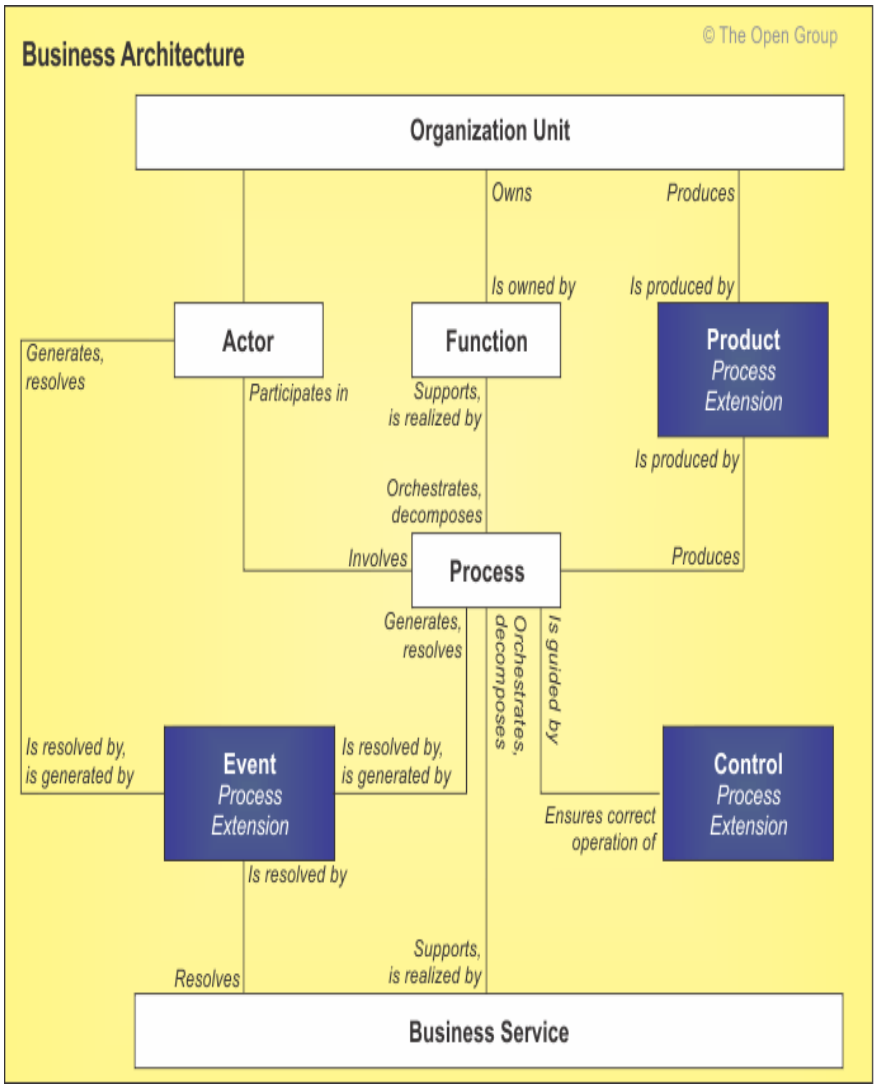
- Scope
  - **Creation of IS services** as an extension of business service
- Additional diagrams to be created
  - Business Use-Case Diagram
  - Organization Decomposition Diagram

Services Extension



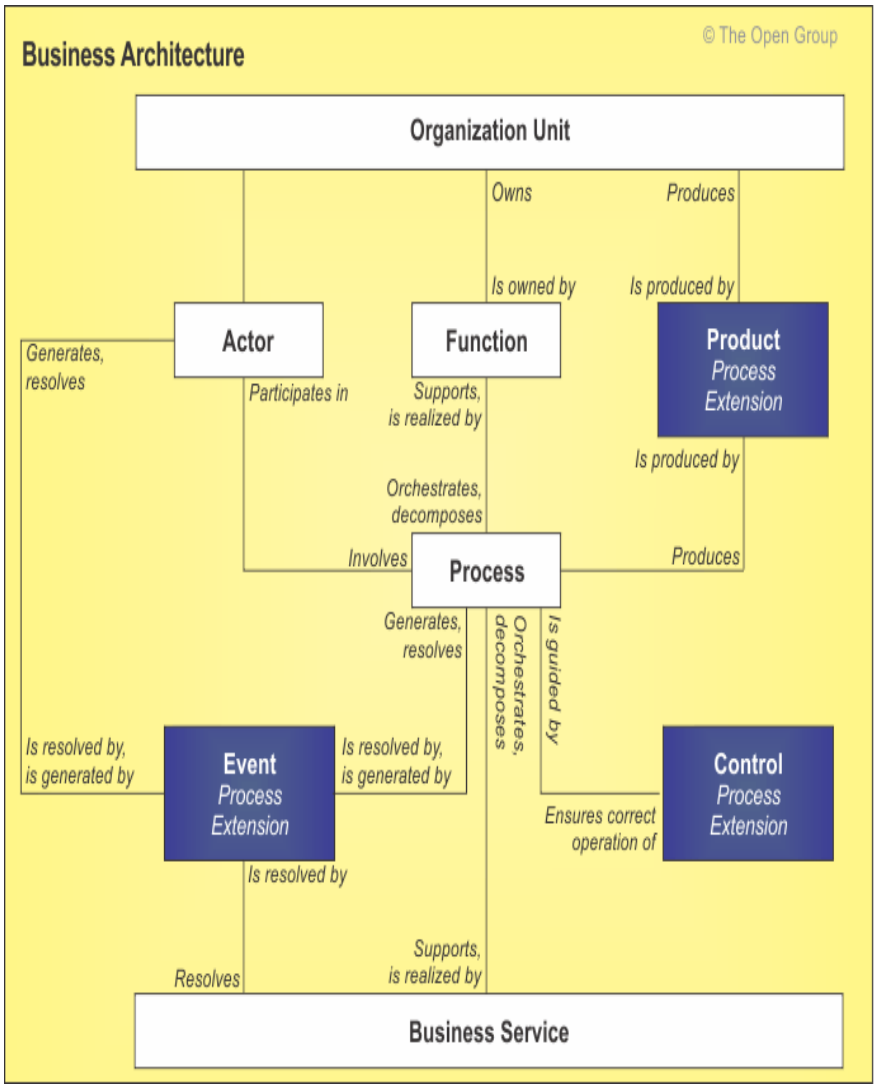
- Should be used
  - When the business has a preset definition of its service that **does not align well** to technical and architectural needs
  - When business and IT use **different language** to describe similar capabilities
  - Where IT service is **misaligned with business need**, particularly around the areas of quality of service, visibility of performance, and management quality
  - Where IT is taking initial steps to **engage business** in discussions about IT architecture

Process Modeling Extension



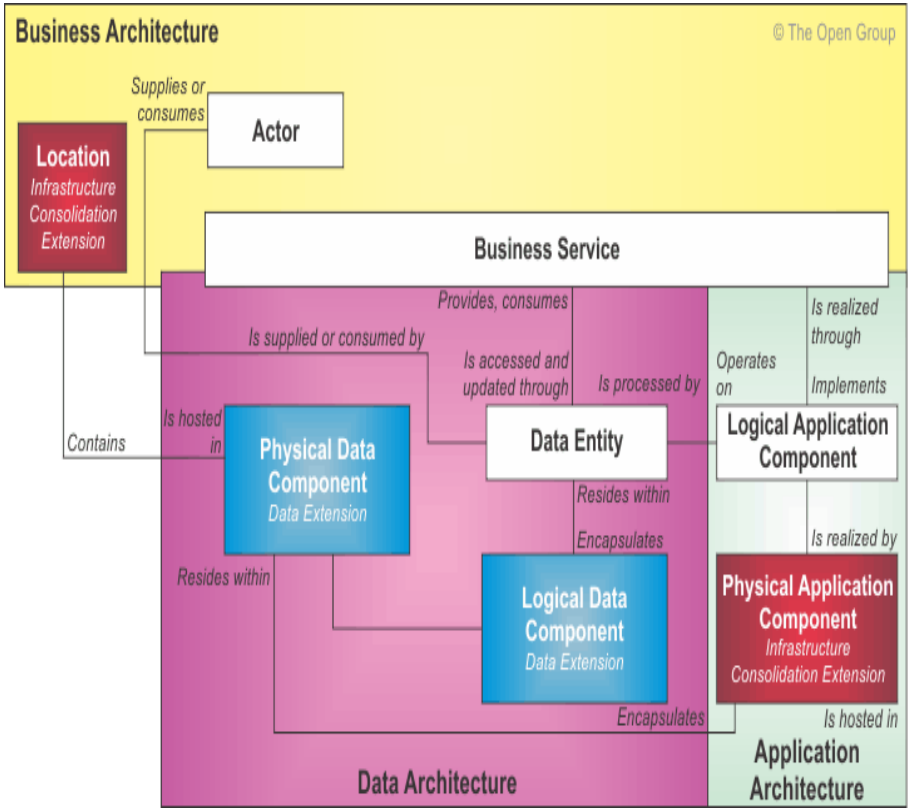
- Scope
  - Creation of **events as triggers** for processes
  - Creation of controls that business logic and governance gates for process execution
  - Creation of products to represent the output of a process
  - Creation of event diagrams to track triggers and state changes across the organization
- Additional diagrams to be created
  - Process Flow Diagrams
  - Event Diagrams

Process Modeling Extension



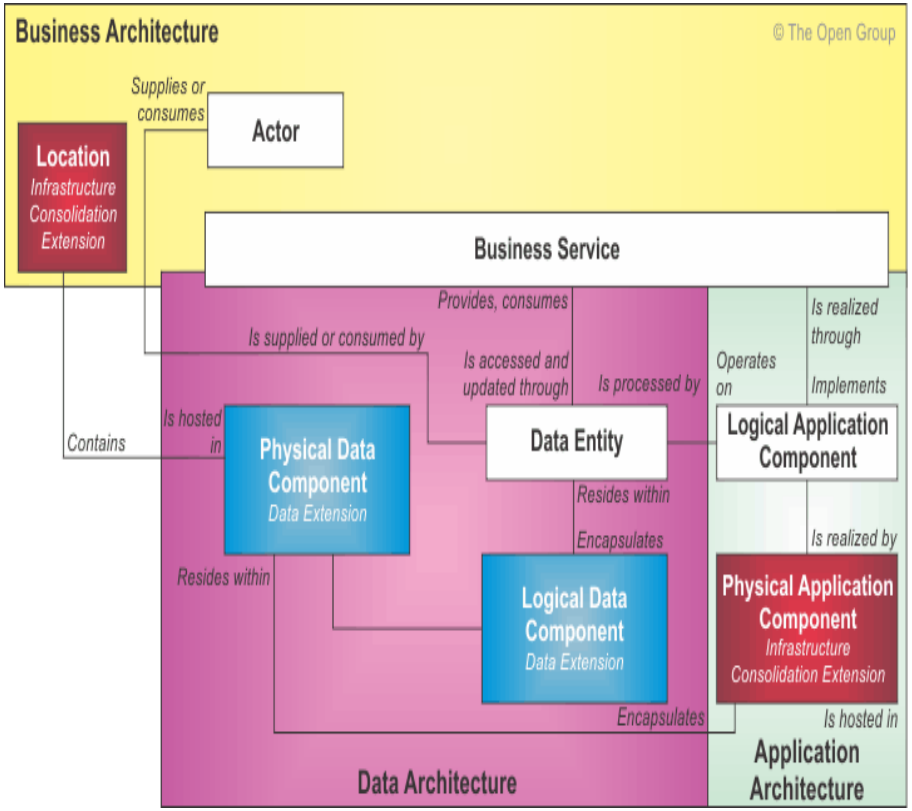
- Should be used
  - Where the architecture must pay specific attention to **state and events**
  - Where the architecture is required to explicitly identify and store process control steps; for example, to support regulatory compliance
  - Where the architecture features critical or elaborate **process flows**

Data Extension



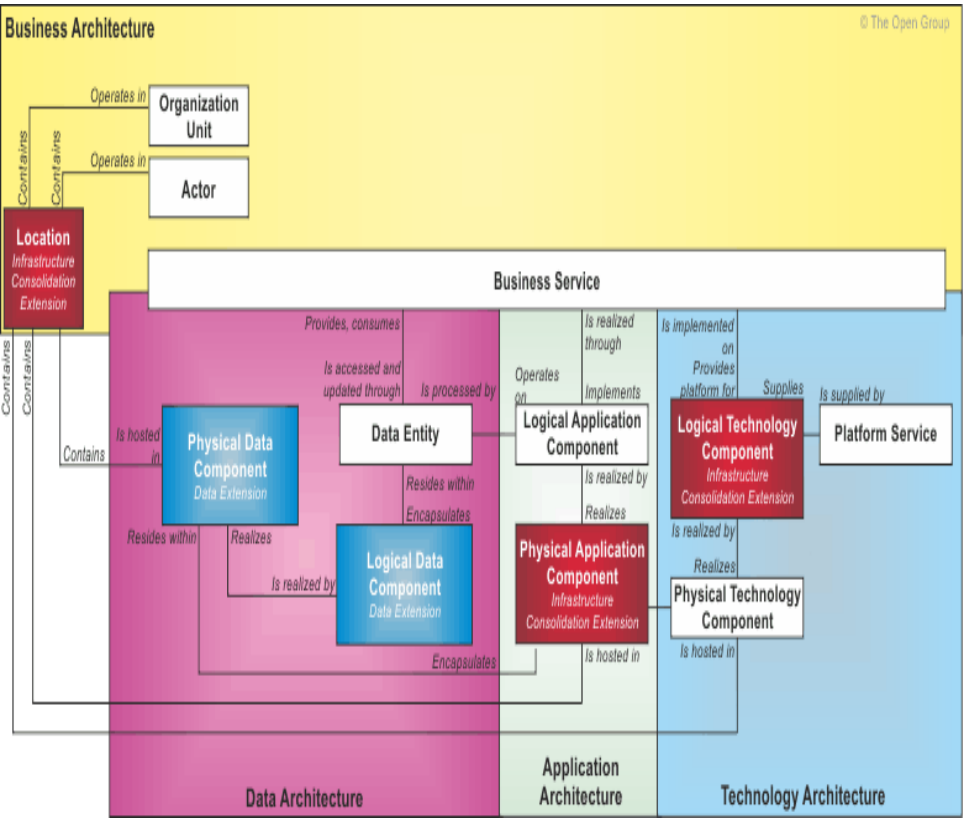
- Scope
  - Creation of **logical data components** that group data entities into encapsulated modules for governance, security, and deployment purposes
  - Creation of **physical data components** that implement logical data components; analogous to databases, registries, repositories, schemas, other techniques of segmenting data
  - Creation of **data lifecycle**, **data security**, and **data migration** diagrams to show data concerns in more detail
- Additional diagrams to be created
  - Data Security Diagram
  - Class Hierarchy Diagram
  - Data Migration Diagram
  - Data Lifecycle Diagram

Data Extension



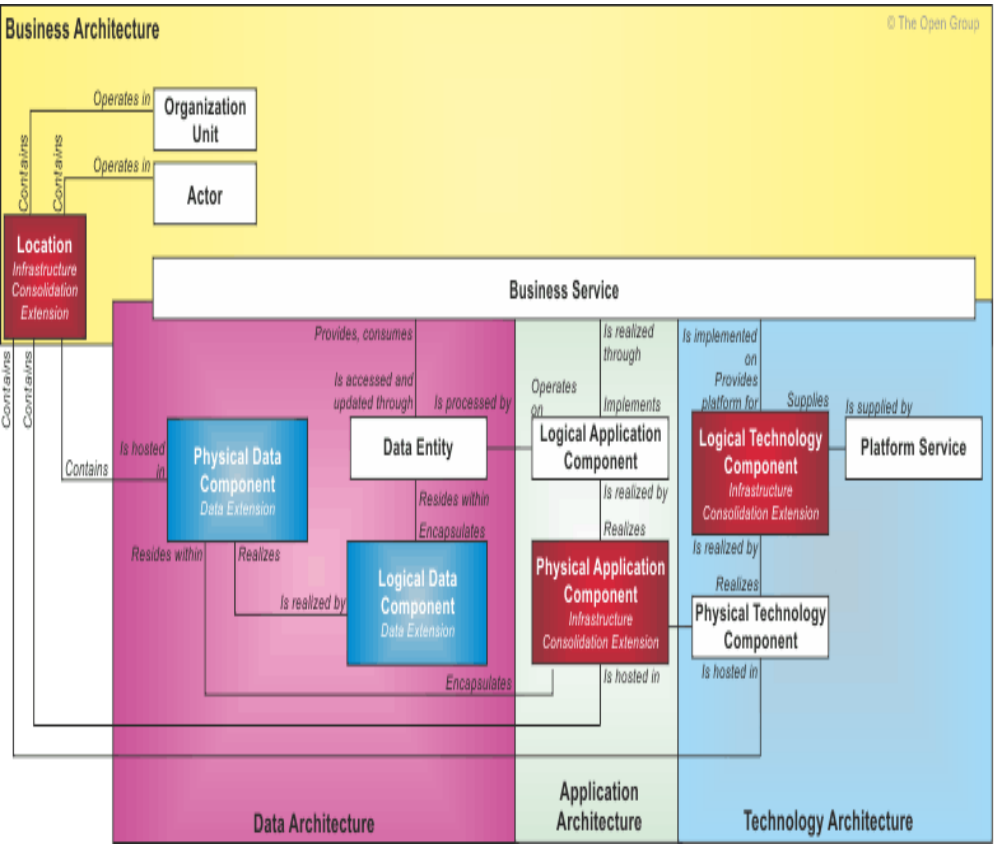
- Should be used
  - Where the architecture features significant complexity and risk around the location, encapsulation, and management of or **access to data**

Infrastructure Consolidation Extension



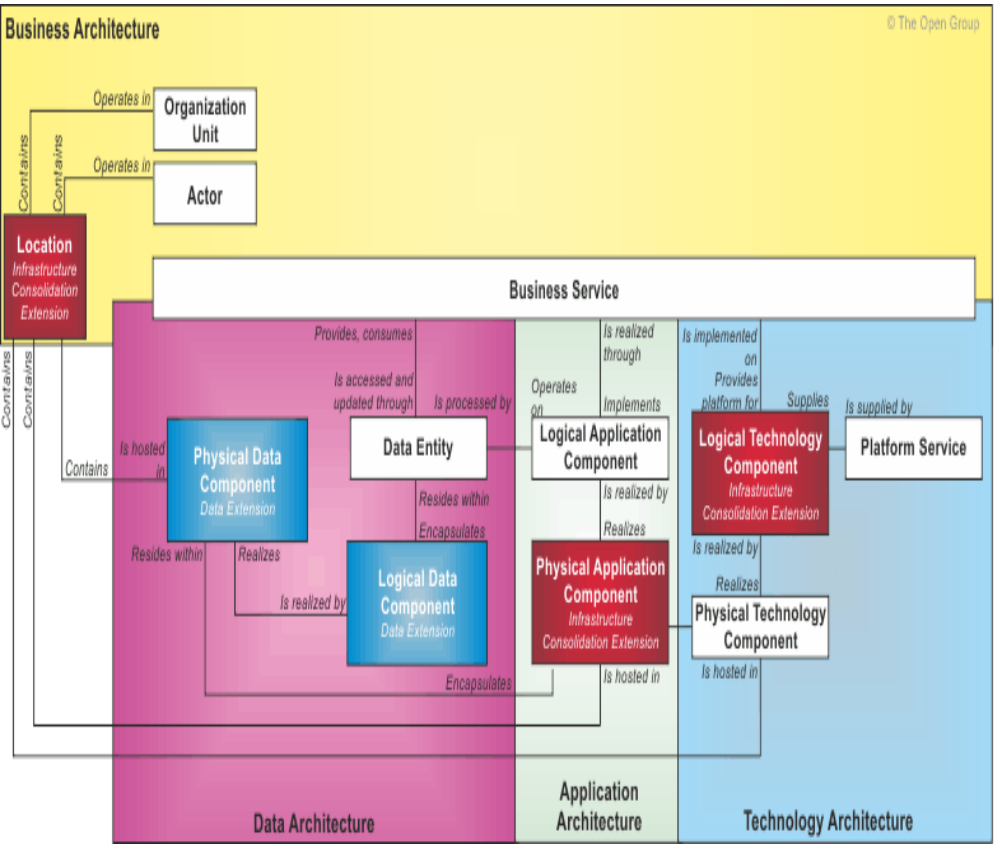
- Scope
  - Creation of a location entity to hold the **location of IT assets** and external consumers or service
  - Creation of **logical and physical application components** to abstract the capability of an application away from the actual applications in existence
  - Creation of logical and physical application components to abstract product type from the actual technology products in existence
  - Creation of additional diagrams focusing on the location of assets, **compliance with standards**, structure of applications, application migration, and infrastructure configuration

Infrastructure Consolidation Extension



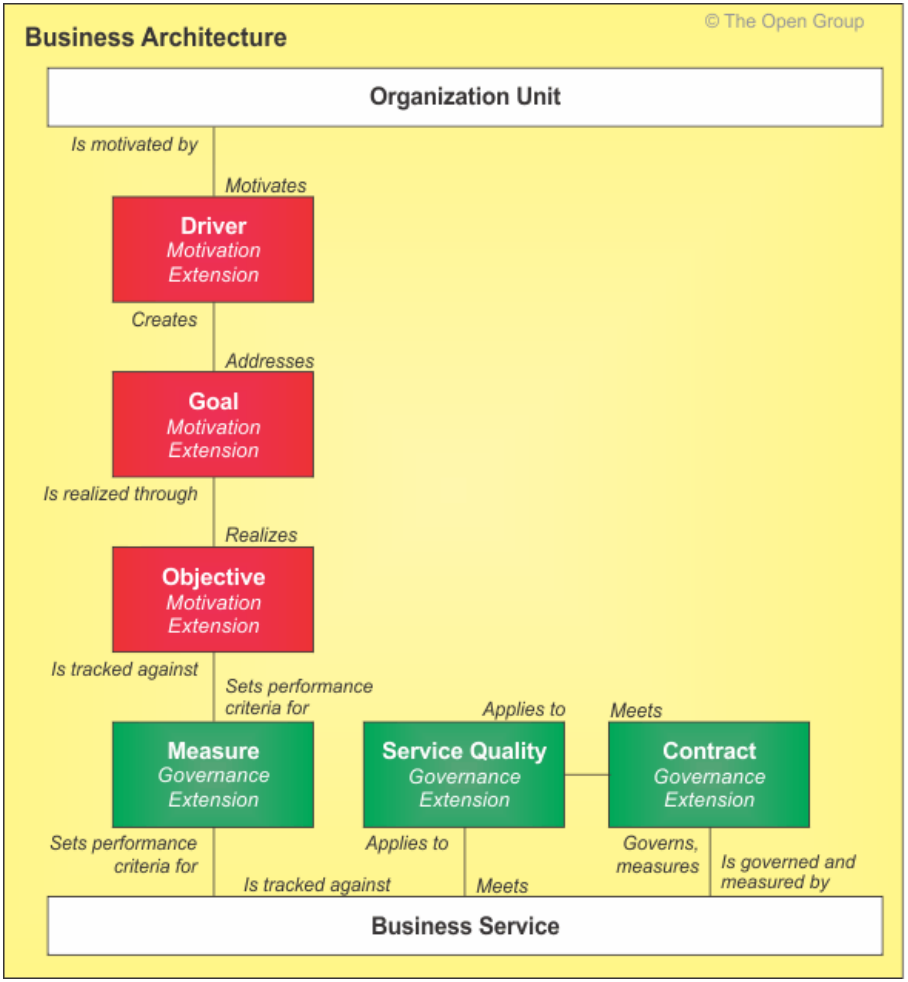
- Additional diagrams to be created
  - Process/System Realization Diagram
  - Software Engineering Diagram
  - Application Migration Diagram
  - Software Distribution Diagram
  - Processing Diagram
  - Networked Computing/Hardware Diagram
  - Communications Engineering Diagram

Infrastructure Consolidation Extension



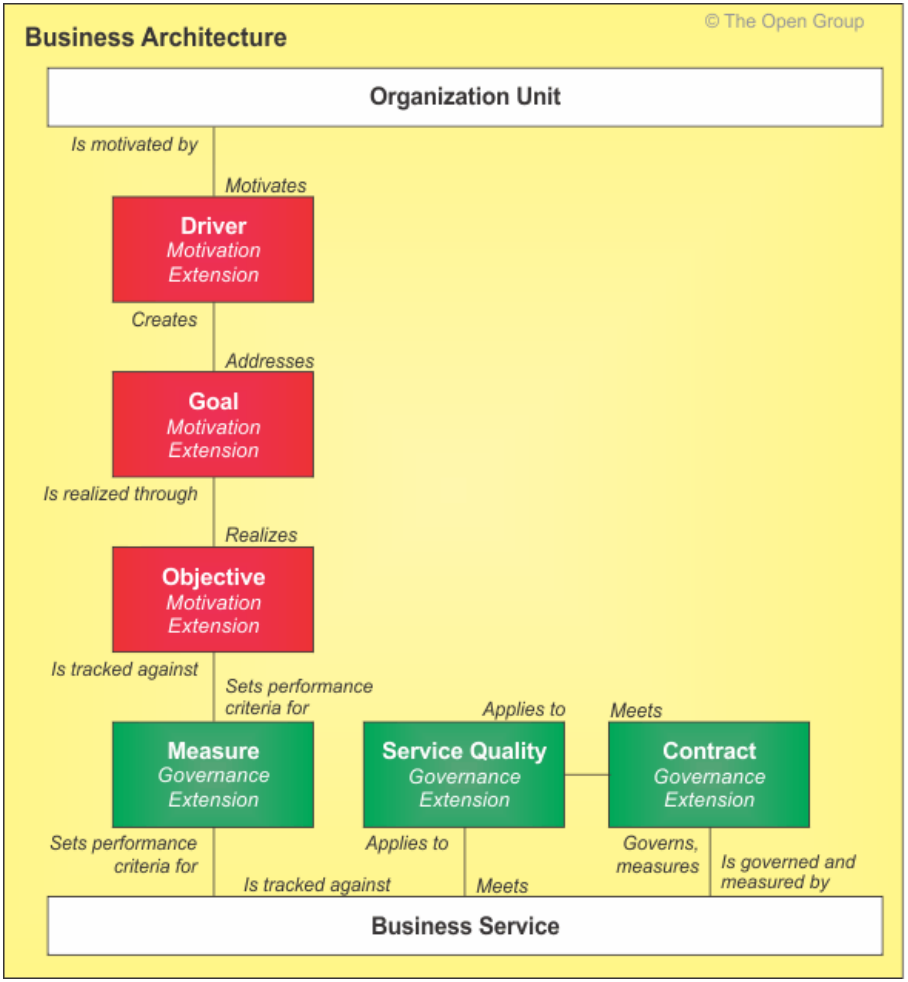
- Should be used
  - Where many technology products are in place with **duplicate** or overlapping capability
  - Where many applications are in place with duplicate or **overlapping functionality**
  - Where applications are geographically dispersed and the decision logic for determining the **location of an application** is not well understood
  - When applications are going to be migrated into a **consolidated platform**
  - When application features are going to be migrated into **a consolidated application**

Motivation Extension



- Scope
  - Creation of a **new metamodel** entity for a **Driver** that shows factors generally motivating or constraining an organization
  - Creation of a new metamodel entity for a **Goal** that shows the strategic purpose and mission of an organization
  - Creation of a new metamodel entity for an **Objective** that shows near to mid-term achievements that an organization would like to attain
  - Creation of a **Goal/Objective/Service** diagram showing the traceability from drivers, goals, and objectives through to services
- Additional diagrams to be created
  - Goal/Objective/Service Diagram

Motivation Extension



- Should be used
  - When the architecture needs to **understand the motivation of organizations** in more detail than the standard business or engagement principles and objectives that are informally modeled within the core content metamodel
  - When organizations have **conflicting drivers** and objectives and that conflict needs to be understood and addressed in a structured form
  - When **service levels are unknown** or unclear

Summary

- TOGAF provides a rich metamodel
- This provides a number of benefits
  - It supports both formal and informal modeling
  - It formalizes the definition of an EA
  - It formalizes the relationship between objects
  - It enables an EA tool mapping